

# 2

## LOGIC HANDBOOK

### 2.1 Introduction

| $p$ | $\neg p$ |
|-----|----------|
| T   | F        |
| F   | T        |

| $p$ | $q$ | $p \wedge q$ | $p$ | $q$ | $p \vee q$ |
|-----|-----|--------------|-----|-----|------------|
| T   | T   | T            | T   | T   | T          |
| T   | F   | F            | T   | F   | T          |
| F   | T   | F            | F   | T   | T          |
| F   | F   | F            | F   | F   | F          |

Let  $p$  and  $q$  be propositions. The conditional statement  $p \rightarrow q$  is the proposition “if  $p$ , then  $q$ .”

The conditional statement  $p \rightarrow q$  is false when  $p$  is true and  $q$  is false, and true otherwise. In the conditional statement  $p \rightarrow q$ ,  $p$  is called the hypothesis (or *antecedent* or *premise*) and  $q$  is called the conclusion (or *consequence*).

“if  $p$ , then  $q$ ”

“ $p$  implies  $q$ ”

“if  $p$ ,  $q$ ”

“ $p$  only if  $q$ ”

“ $p$  is sufficient for  $q$ ”

“a sufficient condition for  $q$  is  $p$ ”

“ $q$  if  $p$ ”

“ $q$  whenever  $p$ ”

“ $q$  when  $p$ ”

“ $q$  is necessary for  $p$ ”

“a necessary condition for  $p$  is  $q$ ”

“ $q$  follows from  $p$ ”

“ $q$  unless  $\neg p$ ”

“ $q$  provided that  $p$ ”

| $p$ | $q$ | $p \rightarrow q$ |
|-----|-----|-------------------|
| T   | T   | T                 |
| T   | F   | F                 |
| F   | T   | T                 |
| F   | F   | T                 |

| $p$ | $q$ | $p \leftrightarrow q$ |
|-----|-----|-----------------------|
| T   | T   | T                     |
| T   | F   | F                     |
| F   | T   | F                     |
| F   | F   | T                     |

Let  $p$  and  $q$  be propositions. The biconditional statement  $p \leftrightarrow q$  is the proposition “ $p$  if and only if  $q$ ”. The biconditional statement  $p \leftrightarrow q$  is true when  $p$  and  $q$  have the same truth values, and is false otherwise. Biconditional statements are also called bi-implications.

| Operator          | Precedence |
|-------------------|------------|
| $\neg$            | 1          |
| $\wedge$          | 2          |
| $\vee$            | 3          |
| $\rightarrow$     | 4          |
| $\leftrightarrow$ | 5          |

A compound proposition that is always true, no matter what the truth values of the propositional variables that occur in it, is called a tautology. A compound proposition that is always false is called a contradiction. A compound proposition that is neither a tautology nor a contradiction is called contingency.

→ Every contingency is satisfiable, but reverse need not to be true 1.

→ Every tautology is satisfiable, but reverse need not to be true.

For any primitive statements  $p, q, r$ , any tautology ‘T’ and any contradiction ‘F’.

(1)  $\neg \neg p \Leftrightarrow p$  Law of Double Negation

(2)  $\neg (p \vee q) \Leftrightarrow \neg p \wedge \neg q$  DeMorgan’s Laws

$\neg (p \wedge q) \Leftrightarrow \neg p \vee \neg q$

(3)  $p \vee q \Leftrightarrow q \vee p$  Commutative Laws

$p \wedge q \Leftrightarrow q \wedge p$

(4)  $p \vee (q \vee r) \Leftrightarrow (p \vee q) \vee r$  Associative Laws

$p \wedge (q \wedge r) \Leftrightarrow (p \wedge q) \wedge r$

(5)  $p \vee (q \wedge r) \Leftrightarrow (p \vee q) \wedge (p \vee r)$  Distributive Laws

$p \wedge (q \vee r) \Leftrightarrow (p \wedge q) \vee (p \wedge r)$

(6)  $p \vee p \Leftrightarrow p$  Idempotent Laws

$$p \wedge p \Leftrightarrow p$$

$$(7) p \vee F \Leftrightarrow p$$

Identity Laws

$$p \vee T \Leftrightarrow p$$

$$(8) p \vee \neg p \Leftrightarrow T$$

Inverse Laws

$$p \wedge \neg p \Leftrightarrow F$$

$$(9) p \vee T \Leftrightarrow T$$

Domination Laws

$$p \wedge F \Leftrightarrow F$$

$$(10) p \vee (p \wedge q) \Leftrightarrow p$$

Absorption Laws

$$p \vee (p \wedge q) \Leftrightarrow p$$

$$p \rightarrow q \equiv \neg p \vee q$$

$$p \rightarrow q \equiv \neg q \rightarrow \neg p$$

$$p \vee q \equiv \neg p \rightarrow \neg q$$

$$p \wedge q \equiv \neg (p \rightarrow \neg q)$$

$$\neg (p \rightarrow q) \equiv p \wedge \neg q$$

$$(p \rightarrow q) \wedge (p \rightarrow r) \equiv p \rightarrow (q \wedge r)$$

$$(p \rightarrow r) \wedge (q \rightarrow r) \equiv (p \vee q) \rightarrow r$$

$$(p \rightarrow q) \vee (p \rightarrow r) \equiv p \rightarrow (q \vee r)$$

$$(p \rightarrow r) \vee (q \rightarrow r) \equiv (p \wedge q) \rightarrow r$$

$$p \leftrightarrow q \equiv (p \rightarrow q) \wedge (q \rightarrow p)$$

$$p \leftrightarrow q \equiv \neg p \leftrightarrow \neg q$$

$$p \leftrightarrow q \equiv (p \wedge q) \vee (\neg p \wedge \neg q)$$

$$\neg (p \leftrightarrow q) \equiv p \leftrightarrow \neg q$$

| Rule of Inference                                                         | Related Logical Implications                                                 | Name of Rule                      |
|---------------------------------------------------------------------------|------------------------------------------------------------------------------|-----------------------------------|
| $p$ $1) \frac{p \rightarrow q}{\therefore q}$                             | $[p \wedge (p \rightarrow q)] \rightarrow q$                                 | Rule of Detachment (Modus Ponens) |
| $p \rightarrow q$ $2) \frac{q \rightarrow r}{\therefore p \rightarrow r}$ | $[(p \rightarrow q) \wedge (q \rightarrow r)] \rightarrow (p \rightarrow r)$ | Law of the syllogism              |
| $p \rightarrow q$ $3) \frac{\neg q}{\therefore \neg p}$                   | $[(p \rightarrow q) \wedge \neg q] \rightarrow \neg p$                       | Modus Tollens                     |